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From Sea to Soil: Marine Actinobacteria in Integrated Seawater Energy Agriculture Systems (ma-ISEAS)

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ABSTRACT

Marine actinobacteria are gaining attention for their biotechnological and ecological potential, yet their roles in marine agriculture, biofuel production, and climate change mitigation remain underexplored. In this review, we examine their application in saline ecosystems—particularly mangroves and *Salicornia*—as biofertilizers and bioinoculants that can enhance crop productivity under extreme conditions. We highlight the ecological functions and secondary metabolites of marine actinobacteria and discuss their potential in sustainable energy production from non-arable lands. We also introduce the concept "marine actinobacteria in Integrated Seawater Energy Agriculture System (ma-ISEAS)" to strengthen the performance of the established ISEAS framework in the United Arab Emirates, thereby supporting renewable biofuel production and contributing to climate change mitigation strategies. Despite this promise, several challenges can be addressed. These include technological bottlenecks in large-scale cultivation, gaps in interdisciplinary collaboration, and the absence of standardized protocols for field application. Furthermore, regulatory and biosafety considerations, especially for the environmental release of genetically enhanced microbial strains, remain underdeveloped. Addressing these challenges is essential for the responsible deployment of marine actinobacteria in advancing agriculture, energy, and environmental sustainability.

1 | Introduction

In the quest for a resilient and equitable future, the intricate interplay between global change, energy security, and sustainable agriculture emerges as a critical focal point (Raza et al. 2025). The urgency to reassess our approaches to energy and food production is underscored by the rapid shifts in climate patterns, resource scarcity, and population growth (Telo da Gama 2023; Javan et al. 2024). Addressing these global challenges requires an interdisciplinary strategy that bridges agricultural science, renewable energy, and environmental systems thinking (Shivanna 2022; Banna et al. 2023; Osuma and Yusuf 2025).

Sustainable agriculture—grounded in resource conservation, greenhouse gas mitigation, and enhanced productivity can—benefit from integration with renewable energy technologies and circular bioeconomy models (Bathaei and Štreimikienė 2023; Kabato et al. 2025). Recent commitments made at COP28, especially during the World Climate Action Summit (WCAS), further highlight the need for coordinated actions that simultaneously reduce emissions, protect nature, and diversify energy systems (United Nations Climate Change, www.unfccc.int/cop28).

This review positions marine actinobacteria as promising agents for this integration. With their remarkable ability to produce bioactive metabolites, degrade organic pollutants, and promote plant growth, marine actinobacteria lie at the intersection of microbial biotechnology, environmental engineering, and sustainable development (Lewin et al. 2016; Shivlata and Satyanarayana 2017; Pattapulavar and Christopher 2025). Their

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applications span carbon cycling, bioremediation, biofuel production, and halophyte-based agriculture, particularly in saline coastal environments (Adegboye et al. 2021; James et al. 2023; Alshamsi et al. 2024; Nazreen et al. 2025).

Coastal ecosystems, including mangroves and *Salicornia*, offer additional leverage. These halophytes not only provide environmental services—such as water filtration and shoreline stabilization—but also serve as viable biomass sources for saline agriculture and sustainable bioenergy (Garza-Torres et al. 2020; Jordan and Fröhle 2022; Ekanayake et al. 2023; Kafeel 2025). However, these ecosystems face growing threats from urbanization and environmental degradation, reinforcing the urgency of nature-based, microbe-assisted resilience strategies (Lyra et al. 2021; Akram et al. 2023).

By introducing the concept of marine actinobacteria in Integrated Seawater Energy Agriculture Systems (ma-ISEAS)—building on the success of the pioneering ISEAS model in the United Arab Emirates (UAE) (www.ku.ac.ae/facilities/seawater-energy-and-agriculture-system; AlYammahi et al. 2024)—this review explores a cross-sectoral innovation platform where saltwater, microbial life, solar energy, and non-arable lands or groundwater converge to support biofuel production, climate adaptation, and sustainable food systems. This holistic model reflects a new frontier for interdisciplinary research and application, uniting microbial ecologists, agronomists, climate scientists, and engineers in advancing sustainable coastal economies.

2 | Actinobacteria in Marine Systems: Characteristics, Biodiversity, and Ecological Roles

Actinobacteria are ubiquitous Gram-positive filamentous bacteria with more than 50% guanine and cytosine content found in their DNA (Dhakal et al. 2017). Members of actinobacteria range from unicellular organisms to filamentous and spore-forming lineages. They reproduce by binary fission or by producing conidia/spores, where sporulation is through fragmentation and segmentation/conidia development (Barka et al. 2016). The majority of actinobacteria produce aerobic substrate and aerial mycelium, while some genera produce only substrate mycelium. The substrate mycelium differs in shape, size, thickness, and colors, ranging from white to yellow, brown, red, pink, orange, green, or black (Holt et al. 1994). Aerial mycelia are thicker than substrate mycelia and differ in the concentric zones, pigmentation, and structure (cotton, velvet, or powder). The morphological appearance is the primary characteristic for the recognition of actinobacterial isolates (Anandan et al. 2016). On culture media, actinobacteria have dry surfaces, and their colonies are compact, tough, or leathery, mostly showing a flat or conical shape with a smooth to slightly wrinkled surface (Barka et al. 2016).

Members of actinobacteria can be recovered from terrestrial, aquatic, coastal salt marshes, and marine habitats (Helmi 2025). Scientists have affirmed their presence in extreme environments, such as acid mines, caves, deserts, and polar regions. They can also exist in symbiotic association with plants, insects, animals, corals, and many fish species (Diarra et al. 2024). The vast majority of actinobacteria are saprophytes capable of producing enzymes that decompose a wide range of plant and animal debris, thereby regulating organic matter (OM) turnover and the carbon cycle (Bhatti et al. 2017; Boubekri et al. 2022).

Actinobacteria offer considerable agricultural benefits, notably through their capacity to produce plant growth-promoting (PGP) compounds (Alshamsi et al. 2024) and act as biocontrol agents (BCAs). These traits are particularly valuable under abiotic stresses, positioning actinobacteria as key players in sustainable terrestrial and marine farming systems. Further specialized roles are explored in Sections 4 and 5.

Marine actinobacteria are considered a "storehouse" of secondary metabolites. About 45% of all bioactive microbial metabolites discovered have been isolated from actinobacteria (Bérdy 2005). The family Actinomycetaceae, including the genera *Streptomyces*, *Micromonospora*, and others, is known for its capability to produce novel bioactive molecules (ul Hassan and Shaikh 2017; Ngamcharungchit et al. 2023). Some strains of actinobacteria have the genetic potential to produce 15–25 secondary metabolites (Lam 2006).

3 | Marine Actinobacteria as Potential Sources of Sustainable Natural Products

Due to their diverse metabolic capabilities and production of bioactive compounds, marine actinobacteria offer innovative solutions in medicine, biotechnology, environmental management, and marine resource utilization. Here are some notable applications of marine actinobacteria:

3.1 | Drug Discovery From Marine Actinobacteria

Marine actinobacteria are a valuable source of novel bioactive compounds with pharmaceutical and biotechnological potential (Lee et al. 2018; Stennett et al. 2020; Pan et al. 2025). Actinobacteria, particularly the genus *Streptomyces*, have been the source of essential antibiotics, including streptomycin, erythromycin, and tetracycline (Hopwood 2007; Lee et al. 2014). These antibiotics have revolutionized medicine by treating bacterial infections and saving countless lives.

Marine actinobacteria can also produce anti-inflammatory agents, immunosuppressants, and other medically related compounds; thus making them valuable for drug development (Jayaprakashvel 2012; De La Hoz- Romo et al. 2025). Some extremophile actinobacteria can produce bioactive molecules adapted to their extreme environments (e.g., deep-sea hydrothermal vents and cold seeps) (Giordano 2020; Hui et al. 2021). *Streptomyces olivaceus* OUCLQ19-3 has the ability to produce a wide array of bioactive secondary metabolites, such as dixiamycins (Cong et al. 2022), which exhibit substantial antibacterial effects against various multidrug-resistant strains (Jin et al. 2021). They have been investigated for the discovery of new antibiotics, antivirals, anticancer agents, and immunomodulators, addressing the growing challenge of drug-resistant pathogens (Busi and Pattnaik 2018; Harikumar and Krishanan 2022). This can be of interest in drug discovery for targeting specific diseases or cellular processes, aligned with sustainability goals.

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3.2 | Environmental Remediation and Bioremediation Potential

Marine actinobacteria are used as bioindicators for assessing the health of marine ecosystems (Soni et al. 2025). Changes in their populations can signal pollution levels and environmental disturbances (Datta et al. 2020; Behera and Das 2023). They also play a crucial role in the bioremediation of marine ecosystems by producing enzymes and secondary metabolites to break down and detoxify organic pollutants in marine ecosystems (Bala et al. 2022; Soni et al. 2025). In this context, marine actinobacteria have been studied for their capacity to degrade various environmental contaminants, such as hydrocarbons, heavy metals, and pesticides (Chaturvedi and Khurana 2019; Mawang et al. 2021). Their enzymatic and metabolic diversity allows them to adapt to different pollutant types and conditions, making them valuable candidates for bioremediation. It has been reported that *Streptomyces* spp. M7 and *Rhodococcus jialingiae* djl-6–2 are involved in wastewater treatment by removing organochlorine and carbamate pesticides, respectively (Benimeli et al. 2006; Xu et al. 2007). Bioremediation using marine actinobacteria holds promise for the restoration and preservation of the health and biodiversity of our oceans and coastal areas.

3.3 | Antifouling Agents and Biocontrol Applications

Marine actinobacteria have emerged as promising candidates for combating biofouling (Canellas et al. 2025) and serving as BCAs in marine environments (Panda and Das 2024). Biofouling is the undesirable accumulation of micro- and macroorganisms on water-submerged surfaces, resulting in increased fuel consumption, maintenance costs, and the spread of invasive species (Magin et al. 2010). Such marine actinobacteria produce secondary metabolites with potent antibiofouling properties that can interfere with the attachment and growth of fouling organisms and prevent their adhesion to surfaces (Canellas et al. 2025). Lobocompactol, derived from *Streptomyces cinnabarinus,* and 6-benzyl and 6-isobutyl 2,5-diketopiperazine derivatives from *Streptomyces praecox* have shown antifouling activity against the bacterium *Pseudomonas aeruginosa*, the diatom *Navicula annexa*, and the macroalgae *Ulva pertusa* (Bavya et al. 2011; Cho et al. 2012).

In addition, 2,5-diketopiperazines and 12-methyltetradecanoid acid from *Streptomyces* sp. inhibited *Balanus amphitrite* and *Hydroides elegans* larval attachment, respectively (Xu et al. 2009, 2010). Napyradiomycins obtained from the marine-derived *Streptomyces aculeolatus* revealed antibiofouling activity against the marine biofilm-forming bacteria and *Mytilus galloprovincialis* larvae (Pereira et al. 2020). Therefore, researchers are exploring the use of extracts or isolated compounds from marine actinobacteria to develop eco-friendly antibiofouling coatings for ships and marine structures and reduce the ecological and economic impact of biofouling.

In addition, marine actinobacteria can act as BCAs by actively inhibiting the proliferation of detrimental pathogens and disease-causing microorganisms in soil and aquatic ecosystems (Panda and Das 2024). Kasugamycin, polyoxin B and D, mildiomycin, and validamycin A are examples that inhibit fungal protein biosynthesis without affecting mammals (Umezawa et al. 1965; Endo and Misato 1969; Feduchi et al. 1985; Barka et al. 2016). Actinomycin D from the marine *Streptomyces parvulus* demonstrated a broad-spectrum activity against the fungal phytopathogens *Magnaporthe grisea*, *Fusarium oxysporum*, *Valsa mali*, and *Rhizoctonia solani* (Yin et al. 2019).

In aquaculture, certain marine actinobacteria exhibit remarkable capabilities to produce antimicrobial compounds to effectively suppress the growth of pathogens (Nazreen et al. 2025). *Nocardiopsis alba* MCCB 110, for example, has the ability to manage diseases associated with the bacterial pathogen *Vibrio harveyi* in fish and shrimp (Sunish et al. 2021). Actinobacteria isolated from marine sediment samples also possess antibacterial activity against different fish pathogenic bacteria belonging to *Vibrio, Aeromonas*, and *Pseudomonas* species (Thirumurugan and Vijayakumar 2013). Researchers aim to develop safer and more sustainable approaches to controlling disease outbreaks in aquaculture settings, reducing the reliance on chemical antibiotics and promoting the health of farmed aquatic species by investing in marine actinobacteria.

3.4 | Functional Roles in Food Preservation and Cosmetics

Extracts and compounds from marine actinobacteria are used in food preservation (Díez et al. 2025) and the formulation of cosmetics (Chinnappa et al. 2024). They produce secondary metabolites, mainly antimicrobial peptides and enzymes, which can inhibit the growth of spoilage microorganisms and foodborne pathogens (Díez et al. 2025). Wijaya et al. (2023) have identified several marine actinobacteria isolated from aquatic environments in Indonesia for their effective antibacterial activities against many foodborne pathogens, such as *Vibrio cholerae*, *Salmonella*, and *Pseudomonas aeruginosa*. Thus, this may offer a safer alternative to chemical preservatives, addressing consumer demands for clean-label and natural food products. Proteases and lipases are among the enzymes produced by marine actinobacteria that improve the texture and flavor of various food products (Tan et al. 2016; Jagannathan et al. 2021).

In addition, marine-derived actinobacteria produce bioactive molecules that possess antioxidant, anti-aging, and moisturizing properties. These compounds can be incorporated into skincare products such as creams, serums, and lotions to enhance their efficacy and appeal to consumers seeking natural and marine-inspired cosmetics. Benzoate and benzoic acid produced from *Streptomyces maritimus* can be used in cosmetics, food preservatives, and hygiene products (Noda et al. 2012; Del Olmo et al. 2017). Thus, this highlights the role of marine actinobacteria in the development of sustainable products in the cosmetics and food sectors.

4 | Marine Actinobacteria in Coastal Agriculture and Aquaculture

4.1 | PGP Roles in Halophyte Agriculture

Marine agriculture plays a critical role in addressing the rising global demand for food, especially in coastal and saline environments (Glenn et al. 1998; Ghirardelli et al. 2024). As

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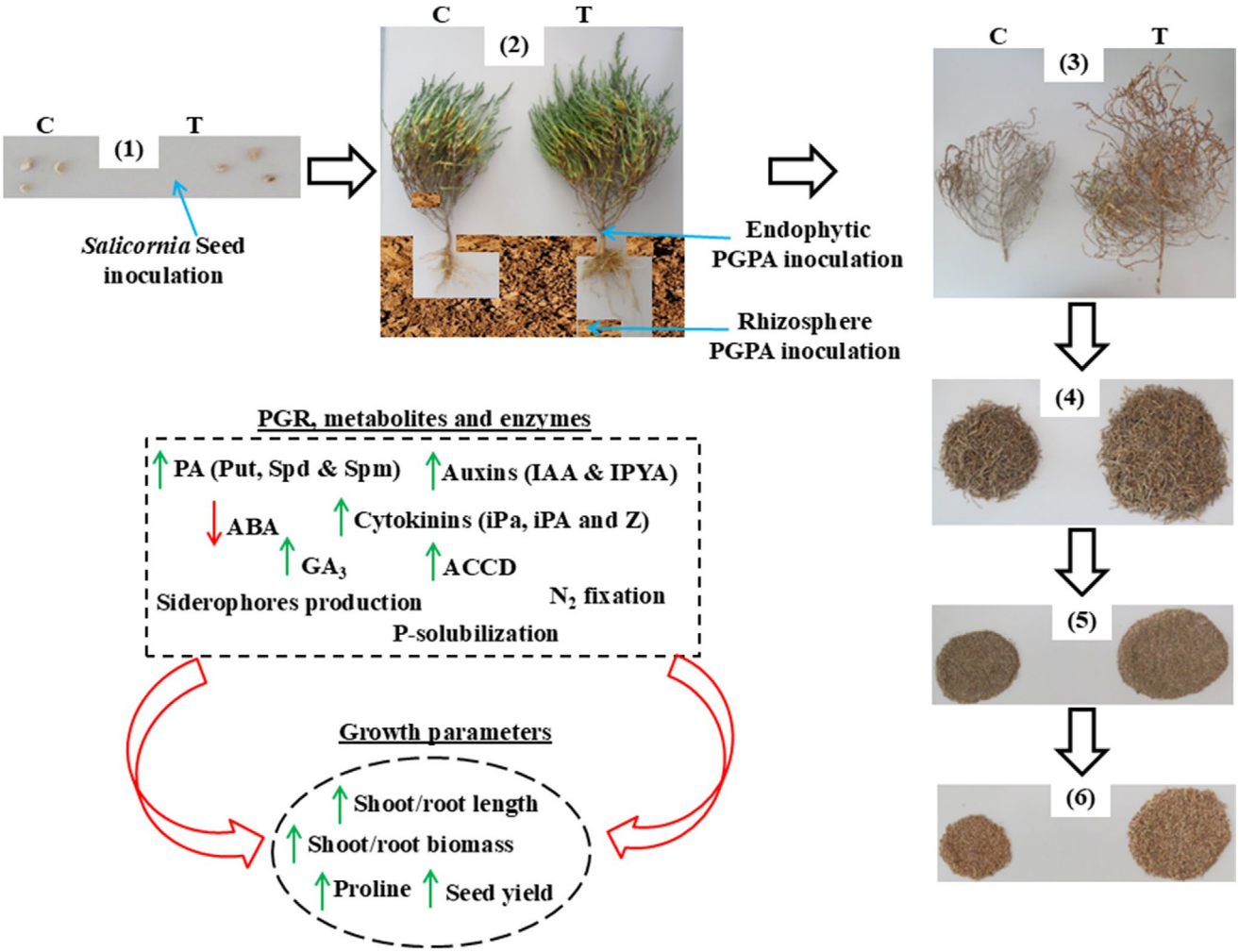

FIGURE 1 | Effects of bio-inoculation of plant growth-promoting actinobacteria (PGPA) on the development stages of *Salicornia*. Stages of growth and development of *Salicornia*: (1) pre-germination treatment and planting stage, (2) seedling development and ongoing stage, (3) vegetative growth (drying and seed set-up) stage, (4) straw stage, (5) seeds and straw stage, and (6) cleaning-up seeds for planting stage. Seeds can be coated/inoculated with PGPA in Stage (1), and seedlings can be inoculated with rhizospheric, rhizoplanic, endorhizospheric, and/or endophytic PGPA during Stage (2) of plant growth. Note the increased growth of PGPA-inoculated *Salicornia* seedlings compared to those non-inoculated (control) seeds/seedlings of *Salicornia bigelovii* (S. AbuQamar and K. El-Tarabily, unpublished). ABA, abscisic acid; ACCD, 1-aminocyclopropane-1-carboxylate deaminase; C, control; CK, cytokinin; DW, dry weight; FW, fresh weight; GA_3, gibberellic acid; IAA, indole-3-acetic acid; iPa, isopentenyl adenine; iPA, isopentenyl adenosine; IPYA, indole-3-pyruvic acid; N_2, nitrogen; P, phosphorus; PA, polyamine; PGR, plant growth regulators; Put, putrescine; Spd, spermidine; Spm, spermine; T, treatment (PGPA inoculation); Z, zeatin.

marine ecosystems face increasing stress from pollution, climate change, and overexploitation, sustainable strategies are needed to support resilient food production systems. Marine actinobacteria offer promising solutions by enhancing the growth of halophytic crops such as *Salicornia* (as illustrated in Figure 1) and improving soil health.

Certain marine actinobacteria, such as *Micromonospora*, facilitate nutrient acquisition and salinity tolerance in *Salicornia*, thereby boosting its productivity in salt-affected soils (El-Tarabily et al. 2019; Mesa-Marín et al. 2023). These actinobacteria can solubilize minerals and fix atmospheric N, making nutrients more bioavailable for plant uptake (Schwob et al. 2018; El-Tarabily, Sham, et al. 2021). In addition, they produce PGP substances like auxins and gibberellins, which stimulate root development and overall plant vigor (Mathew et al. 2020). This symbiotic relationship strengthens *Salicornia*'s capacity to thrive under harsh saline conditions. Similarly, marine actinobacteria support the health and growth of mangroves by breaking down complex OM in the rhizosphere, enhancing nutrient cycling, and aiding in N-fixation (Gupta et al. 2010; El-Tarabily, Ramadan, et al. 2021; Alshamsi et al. 2024). Their contributions are essential for the sustainable cultivation of marine crops and the ecological restoration of degraded coastal zones.

4.2 | Ecological Support and Biocontrol in Aquaculture Systems

Marine aquaculture also benefits significantly from the diverse functions of marine actinobacteria. These microorganisms contribute to the health and stability of aquatic systems by

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improving nutrient availability and supporting beneficial microbial communities (Nazreen et al. 2025).

In coral reef ecosystems, actinobacteria enhance nutrient cycling and help maintain reef health, which is vital for sustaining marine biodiversity and fisheries (Mahmoud and Kalendar 2016). In aquaculture ponds and seaweed farms, they aid in the decomposition of OM, improving water quality and supporting the growth of aquatic plants and animals (Zhou et al. 2023). According to Bull et al. (2005), actinobacteria also facilitate material circulation and the formation of stable soil structures within aquaculture systems.

Marine *Streptomyces* strains, in particular, have shown potential as probiotics in aquaculture. They enhance the digestion and immunity of farmed marine species, such as shrimp and fish, and offer protection against pathogenic bacteria (Ghosh et al. 2023). These benefits contribute to more efficient, eco-friendly practices by reducing the accumulation of waste and mitigating eutrophication in surrounding waters.

Despite these promising applications, the broad adoption of marine actinobacteria in agriculture and aquaculture faces several challenges. These include a limited understanding of their ecological functions, cultivation difficulties, variability in bioactive metabolite production, regulatory issues, and scalability barriers. Addressing these obstacles will require cross-disciplinary collaborations, innovative biotechnological solutions, and supportive policy frameworks to fully harness the potential of marine actinobacteria.

Table 1 provides examples of promising marine actinobacterial strains associated with plant growth promotion, biofuel production, and climate resilience.

5 | Marine Biotechnology and Sustainable Biofuel Production

Marine biotechnology and biofuel production have emerged as critical fields in addressing environmental challenges and energy sustainability (Rai et al. 2025). Marine actinobacteria contribute significantly to marine biotechnology by providing a source of valuable enzymes, bioactive peptides, and bioproducts for the development of novel marine-based technologies (González et al. 2020; Macedo et al. 2021). They also have potential in the sustainable production of biofuels, aligning with the global efforts to reduce reliance on fossil fuels and mitigate environmental impacts (Lewin et al. 2016).

High-quality genome sequencing data obtained from marine actinobacterial strains enable accurate annotation of biosynthetic gene clusters (BGCs) responsible for producing bioactive compounds through homology and domain analysis (Schorn et al. 2016). Comparative genomics approaches from multiple marine actinobacterial strains can identify conserved and variable regions associated with BGCs. Researchers can elucidate the evolutionary history and diversity of biosynthetic pathways across different strains by analyzing gene synteny, sequence similarity, and evolutionary relationships (De Rop et al. 2022). Together, these provide valuable information about the genetic basis of strain-specific metabolic capabilities and the evolutionary mechanisms driving the diversity of bioactive compound production in marine actinobacteria. These genomes may also provide insights into the identification of BGCs associated with secondary metabolite biosynthesis, such as polyketide synthases (PKSs), nonribosomal peptide synthetases (NRPSs), and hybrid PKS–NRPS systems (Doroghazi and Metcalf 2013; Komaki and Tamura 2020). Therefore, researchers can predict the chemical structure and biosynthetic pathways of bioactive compounds, facilitating targeted discovery efforts.

Omics approaches can provide a holistic view of the dynamics and metabolic activities of the microbial community and give insights into the ecological interactions within their native habitats. Metagenomics and metatranscriptomics can also help in the exploration of microbial communities in their natural environments, including marine ecosystems harboring diverse actinobacterial populations, to uncover the genetic potential and expression profiles of BGCs in situ (Mahapatra et al. 2020).

Based on their genomic and proteomic analyses of the cellulolytic enzymes, various actinobacteria can hydrolyze plant biomass into simple sugars and be converted into biofuels and other bioproducts (Wilson 2004; Berlemont and Martiny 2013; Christopherson et al. 2013). Particularly, *Streptomyces*, *Micromonospora*, some strains of *Mycobacterium*, and *Propionibacterium* are enriched in carbohydrate-active enzyme (CAZy) genes, including glycoside hydrolases (GHs) (Lombard et al. 2014). *Streptomyces* sp. can degrade cellulose by endoglucanases, exoglucanases, and β-glucosidase. The plant cell wall polysaccharides xylan, mannan, chitin, hemicellulose, and lignin can be broken down by GHs and/or carbohydrate esterases secreted by various actinobacterial strains. *Actinoalloteichus* sp. and *Streptomyces albidoflavus* isolated from marine sediments, for instance, produced high contents of cellulase and xylanase useful for bioelectricity generation and biofuel production (Fatokun et al. 2016; Gobalakrishnan and Bhuvaneswari 2019; John and Selvarajan 2024).

The marine *Nocardiopsis dassonvillei* PSY13 demonstrated strong cellulase activity that is important in the biofuel industry (Sivasankar et al. 2022). Both *Streptomyces variabilis* and *Streptomyces actuosus*, isolated from the Arabian Gulf sediments and *Mugil cephalus* guts, respectively, also produced cellulase at pH 7.0 and temperatures ranging between 35°C and 37°C (Murugan et al. 2007; Samira et al. 2011). Rajkumar et al. (2021) have shown that the bioconversion of lignocellulosic materials using the cellulose biomass from *Cymodocea* seagrass and cellulase from marine actinobacteria resulted in bioethanol production (John and Selvarajan 2023). The pretreatment of wheat straw and corn cob with the cellulase-producing *Streptomyces* sp. CC48 yielded saccharification, the conversion of complex carbohydrates, such as cellulose or starch, to sugars and ethanol for renewable biofuels (Danso et al. 2022).

The integration of bioinformatics, genomics, and synthetic biology approaches has facilitated genome mining strategies to discover and engineer bioactive compounds in marine actinobacteria. Several cellulase genes were identified in the marine *Streptomyces xinghaiensis* genome mining (Chen et al. 2018).

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TABLE 1 | Potential marine actinobacterial strains associated with growth promotion, biofuel production, and resilience to climate change isolated from marine environments.

Host plant	Actinobacteria species	Mode of action	References
Salicornia bigelovii	*Streptomyces chartreusis* UAE1	Production of auxins	Mathew et al. (2020)
	Streptomyces tritolerans UAE1	Production of PA	
	Streptomyces rochei UAE1	Production of ACCD	
	Actinoplanes deccanensis UAE1	Production of PA	El-Tarabily et al. (2020)
	Streptomyces euryhalinus UAE1	Production of PA	
	Micromonospora chalcea UAE1	Production of ACCD	El-Tarabily et al. (2019)
Mangroves	*Streptomyces mutabilis* UAE1	Production of PA	El-Tarabily, Ramadan, et al. (2021)
	Streptomyces coelicoflavus UAE1	P-solubilization	El-Tarabily, Sham, et al. (2021)
	Streptomyces polychromogenes UAE1	Production of auxins	
	Streptomyces bacillaris UAE1	Production of PA	
	Streptomyces ferrugineus UAE1	Production of ACCD	
	Streptomyces tubercidicus UAE1	Production of PGRs and ACCD, and P-solubilization	Alkaabi et al. (2022)
	Streptomyces griseorubens UAE1	Production of PGRs and ACCD	Alshamsi et al. (2024)
	Streptomyces spp.	Production of auxins and siderophores, nitrate reduction, and P-solubilization	Shrivastava et al. (2015)
	Streptomyces galbus	P-solubilization	Sahu et al. (2007)
	Streptomyces sp.	P-solubilization	Gupta et al. (2010)
	Nocardiopsis yanglingensis	P-solubilization	Suksaard et al. (2017)
	Streptomyces jiujiangensis	Production of auxins	
	Streptomyces psammoticus	Production of ACCD	
	Pseudonocardia oroxyli	Production of siderophore	
Sea lavender (*Limonium sinense*)	*Streptomyces* sp. KLBMP 5084	Genes related to PGP and salt tolerance	Qin et al. (2017)
Agarwood (*Aquilaria crassna*)	*Nocardia alba* GQ179653	Production of IAA, ammonia, and hydroxamate-like siderophore; and protease activity	Nimnoi et al. (2010)
	Nocardia jiangxiensis S3308 and S3311	Production of IAA, ammonia, and hydroxamate-like siderophore, and protease activity.	
	Actinomadura glauciflava GQ179654	Production of IAA, ammonia, and siderophore, and protease activity	
	Nomomuraea rubra GQ179656	Production of IAA, and hydroxamate-like siderophore, and protease activity	
	Streptomyces javensis GQ179657	Production of IAA, ammonia, and hydroxamate-like siderophore, and protease activity	
	Pseudonocardia halophobica GQ179660	Production of IAA and ammonia, and protease activity	

Abbreviations: ACCD, 1-aminocyclopropane-1-carboxylate deaminase; IAA, indole-3-acetic acid; P, phosphorus; PA, polyamine; PGP, plant growth promoting; PGRs, plant growth regulators.

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Due to their ability to synthesize intracellular and extracellular lipids, including triacylglycerols (TAGs) and polyunsaturated fatty acids (PUFAs), marine actinobacteria can be potential candidates for biofuel production (John et al. 2022; Senadheera et al. 2023). These lipids serve as precursors for biodiesel and biofuel production through microbial lipid fermentation and transesterification processes.

Owing to their tolerance to high salinity and temperatures, actinobacteria are considered suitable candidates for biofuel production from marine biomass feedstocks. In general, their metabolic versatility and genetic tractability offer opportunities for metabolic engineering to enhance lipid productivity and optimize biofuel yields. However, limited understanding of their metabolic pathways, genetic manipulation tools, and scalability issues pose significant hurdles hindering the widespread application of marine actinobacteria in biotechnology and biofuel production. In addition, regulatory constraints, intellectual property issues, and economic viability impede the commercialization of marine actinobacteria-derived products. Addressing these challenges necessitates interdisciplinary collaborations, innovative bioprocessing techniques, and policy frameworks conducive to sustainable biotechnological development.

6 | Halophyte Crops in Marine Agriculture Systems

6.1 | Mangroves: Ecosystem Engineers and Carbon Sinks

Mangroves have extensive root systems that help stabilize coastlines, reducing the impact of erosion caused by waves and tidal action (Sunkur et al. 2023). This protective function is crucial for maintaining the integrity of coastal areas where marine agriculture may be practiced. Mangroves can also serve as critical nursery habitats for various marine species, including fish and crustaceans (Friis and Killilea 2024). The complex root systems provide shelter and protection for juvenile organisms, supporting the sustainability of fisheries and aquaculture. These plants are highly effective carbon sinks, sequestering large amounts of carbon dioxide (CO_2) from the atmosphere (Soeprobowati et al. 2024). This not only helps mitigate climate change but also contributes to the health of coastal ecosystems.

6.2 | *Salicornia*: A Bioenergy Crop for Arid and Saline Lands

Salicornia, also known as glassworts or sea asparagus, is a cash crop halophyte with remarkable adaptability to extreme saline environments (Sharma et al. 2016; Garza-Torres et al. 2020; Kafeel 2025). Cultivated in coastal areas where traditional crops struggle, it diversifies marine agriculture by offering a viable alternative. Among its numerous species, *Salicornia bigelovii* stands out for its ability to survive in saltwater-irrigated crops in coastal deserts worldwide (Glenn et al. 1999; Doncato and Costa 2018). Successful cultivation of *Salicornia* has been demonstrated in several countries, including the USA, Mexico, Egypt, Eritrea, Pakistan, Saudi Arabia, and the UAE (Al-Yamani et al. 2013; Doncato and Costa 2018; El-Tarabily et al. 2019).

This resilient plant is not just a survivalist; it excels as a photosynthesizer, outperforming even staple grains like wheat (Watson and Byrne 2009). Its economic significance is substantial, serving as both a bioresource and culinary delight (Loconsole et al. 2019). High yields of foliage and oilseeds make it valuable for animal feed and human consumption. Researchers have also highlighted *Salicornia* oil as a potential source of polyunsaturated fat and biodiesel supply to meet global demands (Glenn et al. 1991; Bailis and Yu 2012). About 30% of its seed content is oil, making it a promising energy crop for large-scale cultivation across various countries (Ekanayake et al. 2023).

The oil extracted from *Salicornia* seeds is not only used for biodiesel production but also yields co-products like seedcake and straw biomass. These byproducts find utility as livestock diet supplements, further enhancing the plant's economic value (Doncato and Costa 2018). Moreover, its succulent stems and salty flavor make it a popular culinary ingredient in many countries, adding zest to salads, seafood dishes, and pickles (Ekanayake et al. 2023).

Beyond its economic and culinary merits, *Salicornia* plays a crucial role in sustainable land management. Its cultivation improves soil health in saline and degraded areas, stabilizing soil, preventing erosion, and contributing to nutrient cycling. In essence, *Salicornia* embodies a holistic approach to agriculture, offering solutions to environmental challenges while enriching our planet.

7 | Marine Agriculture Management

Building on its established benefits, marine agriculture is being integrated into innovative systems like ma-ISEAS to enhance biomass production and sustainability of halophyte-based agriculture.

7.1 | Halophyte–Algae Systems and Resource Optimization

Halophyte agriculture and algae systems, discussed by Hendricks and Bushnell (2009), represent an innovative approach to marine agriculture management, emphasizing the cultivation of salt-tolerant plants and algae in coastal areas. This method enables the natural resilience of halophytes and various species of algae to create sustainable agricultural systems. These systems are particularly important in addressing the limitations of traditional agriculture, which often relies heavily on freshwater resources. By utilizing seawater and marginal lands unsuitable for conventional crops, halophyte agriculture and algae systems can expand arable land and reduce the strain on freshwater supplies.

The significance of these systems lies in their potential to contribute to food security and economic development, especially in regions prone to water scarcity and soil salinization. Halophytes can be cultivated for their edible parts, biofuel production, and as animal fodder, offering versatile uses that can diversify agricultural output (Barkla et al. 2024). Algae, on the other hand, are highly

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efficient at photosynthesis and can produce biomass at a faster rate than terrestrial plants, making them a valuable resource for bioenergy, pharmaceuticals, and as a nutritional supplement (Razzak et al. 2024). The integration of these systems can thus support sustainable agricultural practices while enhancing the resilience of food production systems to environmental stresses.

Furthermore, halophyte and algae systems play a crucial role in environmental conservation and climate change mitigation (Hameed et al. 2024). These plants and algae can sequester CO_2, contributing to the reduction of greenhouse gases in the atmosphere. In addition, halophytes can help stabilize coastal areas by preventing soil erosion and promoting land reclamation. Algae cultivation, particularly in integrated multi-trophic aquaculture systems, can improve water quality by absorbing excess nutrients and pollutants, thus maintaining the health of marine ecosystems (Vijayaram et al. 2024). This dual function of production and environmental protection highlights the importance of marine agriculture management in achieving ecological balance and sustainability. The stages described by Hendricks and Bushnell (2009) can be summarized as the following:

1. **Selection and cultivation:** The halophytes, *Salicornia bigelovii*, *Kosteletzkya virginica*, and *Distichlis spicata*, were selected for their ability to tolerate high salt levels and produce viable yields. They also emphasized using high-yield algae strains rich in lipids or carbohydrates and developing cultivation techniques, such as CO_2-force-fed photobioreactor systems, to enhance growth rates and biomass accumulation. They recommended utilizing bacterial strains capable of producing biofuels or valuable biochemicals and optimizing fermentation or other growth conditions to increase yield.
2. **Harvesting and processing**: Efficient methods for harvesting halophytes using mechanical or manual techniques were developed to process biomass for extracting oils, proteins, or fibers.
3. **Conversion to food and fuel**: Converted extracted oils from halophytes into biofuels using transesterification for biodiesel production and processed their proteins and fibers for food or feed applications. Converted algal lipids into biodiesel or other biofuels through transesterification or other chemical processes, utilized algal carbohydrates for bioethanol production, and processed remaining algal biomass for food or feed products. Converted bacterial metabolites or biomass into biofuels like ethanol, butanol, or hydrogen through fermentation or other conversion processes, and explored additional applications for bacterial products in the food or feed industries.
4. **Optimization and scaling**: They optimized cultivation, harvesting, and processing techniques to improve efficiency and yield. They scaled up the processes from laboratory or pilot scale to commercial production levels and addressed challenges related to economic viability, environmental impact, and sustainability.
5. **Integration and implementation**: Hendricks and Bushnell (2009) integrated the production of halophytes, algae, and bacteria into existing agricultural and industrial systems. They developed strategies for large-scale implementation and commercialization and assessed the overall impact on food security, energy production, and environmental sustainability.

The adoption of halophyte agriculture and algae systems can drive innovation in agricultural technologies and practices. Research and development in this field can lead to the discovery of new crop varieties, improved cultivation techniques, and efficient resource management strategies. Genetic engineering and selective breeding can enhance salt tolerance and productivity of halophytes and algae, making them more viable for large-scale agriculture (Cárdenas-Pérez et al. 2021; Afzal et al. 2023). The knowledge gained from these advancements can also be applied to traditional crops, helping them adapt to changing environmental conditions. Overall, this work highlights the critical role of marine agriculture management in fostering a sustainable and resilient agricultural future.

On the other hand, several limitations can be associated with halophyte agriculture and algae systems. One significant constraint is the economic viability of these alternative agricultural methods. The initial investment and ongoing maintenance costs are considerably high, often outweighing the benefits when compared to conventional agriculture (Abideen et al. 2023). Halophyte crops capable of thriving in saline environments produce lower yields and have limited market acceptance due to unfamiliarity and niche demand (Barkla et al. 2024). Despite their high productivity and potential for biofuel production, algae systems face technical challenges such as efficient harvesting, dewatering, and processing (Gaurav et al. 2024). It has also been emphasized the environmental concerns, such as the risk of salinization of freshwater resources and the potential negative impacts on local ecosystems (Hendricks and Bushnell 2009). While promising, these systems require significant technological advancements and economic adjustments to become mainstream agricultural solutions.

7.2 | The Integrated Seawater Energy Agriculture System (ISEAS): UAE Experience

Since 2014, the International Center for Biosaline Agriculture (ICBA) in the UAE has been evaluating the use of seawater to cultivate *Salicornia* for biofuel production. ICBA has been diligently assessing all existing genotypes of *Salicornia* using agronomic practices and inbred seed lines. Preliminary trials have shown promising results, with fresh biomass and seed yields exceeding expectations. Fresh biomass varies from 40 to 60 tons/ha/year, whereas seed yield varies from 0.5 to 1.9 tons/ha/year, producing four times more seeds than other genetic material (Al-Yamani et al. 2013; Lyra et al. 2016). The work has further validated the oil and seed content (AlYammahi et al. 2024). Khalifa University (KU) in the UAE has also initiated *Salicornia* cultivation for biofuels, with a two-hectare farm in Masdar City. The Sustainable Bioenergy Research Consortium (SBRC) at KU is leading the ISEAS (AlYammahi et al. 2024), which utilizes saltwater and desert land to grow salt-tolerant crops for biofuel production. This system not only supports the UAE's aquaculture

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sector but also aims to bolster the economy by contributing to various sectors, including aviation, oil refining, agriculture, and aquaculture. Researchers envision ISEAS as a potential source of sustainable aviation fuel in the future.

The project, developed by KU-UAE and sponsored by Etihad Airways, Boeing, General Electric, and Honeywell UOP, supports Sustainable Aviation Fuels (SAF) to combat climate change by reducing CO_2 emissions. In February 2018, *Salicornia* oil was processed at Takreer Research Center, UAE, and converted into aviation biofuel. This biofuel was mixed with jet fuel to power an Etihad Airways flight on a Boeing aircraft. In January 2019, Etihad Airways successfully operated the first commercial flight (Boeing 787) from Abu Dhabi to Amsterdam, partially fueled by biofuel derived from *Salicornia* oil, blended directly with jet fuel without modifications to the aircraft, engines, or airport fueling systems. The UAE Energy Plan 2050 aims to cut CO_2 emissions by 70%, increase clean energy use by 50%, and improve energy efficiency by 40%. Although around 160,000 passenger flights globally have used sustainable and traditional jet fuel, Etihad's flight was the first powered by biofuel produced in the desert using halophytic plants irrigated with saltwater.

The ISEAS represents a groundbreaking approach to sustainable resource utilization. It seamlessly integrates aquaculture, halo-agriculture, and mangrove cultivation to produce not only sustainable biofuels for aviation but also a variety of valuable byproducts like seafood. This innovative system capitalizes on the synergies and interactions within integrated desert ecosystems, optimizing resource efficiency and ecosystem health.

At the heart of ISEAS lies its closed-loop design (Figure 2A). Seawater sourced from the Arabian Gulf undergoes a series of stages within the system, culminating in its filtration through mangrove plantations. These mangroves not only purify the water by removing nitrates and ammonia but also serve as carbon sinks, absorbing CO_2 from the air and mitigating the effects of global warming. The water, now cleansed and enriched, can be returned to the Gulf or recycled back into the system for use in fish farms, completing the sustainable cycle. Notably, all energy required for the system's operation, including water pumping, is generated by solar arrays, eliminating the need for fossil fuel input entirely. The map and functions of ISEAS at KU can be found in Figure 2A and outlined in six sequential phases as the following:

1. **Aquaculture initiation:** Seawater is pumped into aquaculture ponds to support the growth of shrimp and fish, ensuring a steady food supply.
2. **Aquaculture expansion:** The aquaculture section comprises six open, lined ponds with a total capacity of 1277 m^3, facilitating the growth and development of aquatic species.
3. **Halo-agriculture integration:** Nutrient-rich wastewater from aquaculture operations is repurposed to irrigate and fertilize the halo-agriculture section, featuring eight *Salicornia* fields. These fields, each spanning 685 m^2, cultivate halophytic plants suitable for arid environments with saltwater irrigation.
4. **Harvesting and utilization:** Halophytic plants are harvested for their oilseeds, with the remaining seed meal serving as feed for fish, shrimp, or a protein source for animal feed, ensuring minimal waste and maximum resource utilization.
5. **Water management and recirculation:** *Salicornia* fields are equipped with perforated pipes to drain excess irrigation water. This water can be recirculated to the irrigation pump or directed toward mangrove swamps as continuous effluent, contributing to water conservation and ecosystem health.
6. **Mangrove incorporation:** The mangrove section consists of four artificial wetlands/swamps and a terminal exit water trench. Mangroves, with their extensive root structures, act as carbon sinks, absorbing CO_2 from the air. In addition, biomass from the mangroves can be converted into bioenergy, further enhancing sustainability.

Fundamentally, ISEAS integrates various components seamlessly, holding synergies between aquaculture, halo-agriculture, and mangrove cultivation to optimize resource use, promote ecosystem health, and contribute to sustainable biofuel production.

7.3 | Marine Actinobacteria in ISEAS (ma-ISEAS): Concept and Applications

The ma-ISEAS enhances the established ISEAS (Section 7.2) by integrating marine plant growth-promoting actinobacteria (PGPA). This system leverages PGPA's metabolic pathways to optimize seawater agriculture using salt-tolerant crops and renewable energy (solar/wind/tidal) while improving resource efficiency through waste recycling and synergistic energy–nutrient flows (Figure 2B).

8 | ma-ISEAS: Sustainability, Economics, and Policy Perspectives

The economic benefits of ma-ISEAS vary by location, scale, technology, and market conditions. However, several potential advantages have been identified:

1. **Energy cost savings:** Technologies such as ocean thermal energy conversion (OTEC) and seawater-air conditioning (SWAC) can significantly reduce energy costs compared to conventional cooling and power systems (Vega 2017; Lucas and Sanjivy 2024).
2. **Agricultural revenue:** The use of seawater and renewable energy for crop cultivation and aquaculture supports the production of high-value outputs such as seafood, algae, and halophyte crops, contributing to revenue generation in coastal areas (Brown et al. 2018).
3. **Enhanced productivity**: Marine actinobacteria contribute to soil fertility and plant health, reducing the need for chemical fertilizers and enhancing yields in saline environments. (Abdellatif et al. 2024).

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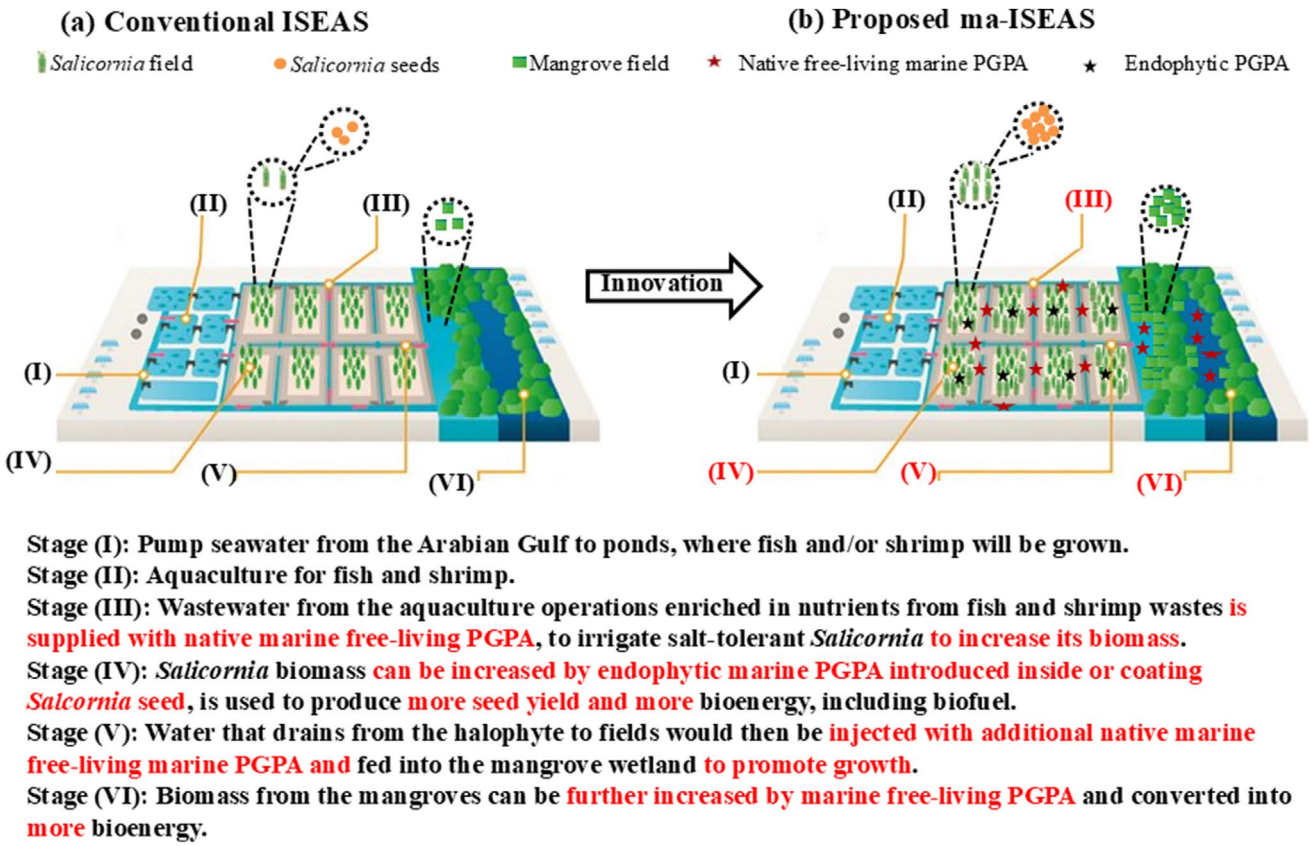

FIGURE 2 | Scheme of the stages of the innovative marine actinobacteria-Integrated Seawater Energy Agriculture System (ma-ISEAS). Comparisons between the stages of the (a) conventional ISEAS, which was originally developed by the Sustainable Bioenergy Research Consortium (SBRC) at Khalifa University (KU), and (b) the proposed ma-ISEAS. Native marine free-living PGPA will be used for both *Salicornia* and mangroves, whereas endophytic PGPA will coat *Salicornia* seeds. Numbers (I–VI) represent the stages of ISEAS (modified from the SBRC; www.ku.ac.ae/facilities/seawater-energy-and-agriculture-system; AlYammahi et al. 2024). Bold red font refers to the modification/improvement done on that particular stage of proposed ma-ISEAS. PGPA, plant growth-promoting actinobacteria.

4. **Job creation:** The development of ma-ISEAS systems can stimulate employment in multiple sectors, including sustainable agriculture, marine biotechnology, renewable energy, and environmental engineering (UNEP 2011).
5. **Environmental savings:** Integrating seawater-based processes and renewable energy reduces reliance on freshwater, cuts carbon emissions, and helps avoid ecological degradation—lowering long-term health and environmental remediation costs (Alnaqbi and Alami 2023).
6. **Niche market opportunities:** The production of sustainable and premium-quality seawater-based products (e.g., bio-cosmetics, sea vegetables, and functional foods) can tap into high-value markets with growing consumer demand for eco-friendly products (Morais et al. 2021; Pereira and Cotas 2024).
7. **Tourism and eco-recreation:** Strategically designed ma-ISEAS facilities can double as ecotourism destinations, offering educational, recreational, and cultural experiences that benefit local economies (Satrya et al. 2023).
8. **Government support:** Public funding, including grants, subsidies, and policy incentives for integrated sustainability solutions (IRENA 2019), can significantly enhance the financial viability of the ma-ISEAS project.
9. **Climate resilience:** Diversification of food and energy sources through ma-ISEAS improves adaptive capacity and reduces economic vulnerability to climate-induced water and energy shortages (Altieri et al. 2015).

While these benefits are promising, comprehensive techno-economic assessments are essential to validate the feasibility of ma-ISEAS in practice. Studies applying life cycle assessments (LCA), cost–benefit analysis (CBA), and integrated resource modeling can provide more accurate evaluations of investment requirements, operational costs, market trends, and socioeconomic impacts (Bruno et al. 2023; Akter et al. 2024). Further research is needed to develop context-specific economic models that support decision-making and policy formulation for the adoption of ma-ISEAS.

9 | Conclusions: A Convergent Pathway Toward Resilient Blue–Green Systems

Marine actinobacteria, with their exceptional metabolic and ecological versatility, offer transformative opportunities at the intersection of microbiology, sustainable agriculture, and renewable energy systems. Their roles in nutrient cycling, plant growth promotion, ecosystem restoration, and biofuel

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production position them as key players in shaping climate-resilient food–energy–water systems. By integrating them into advanced systems like ma-ISEAS, we not only expand the boundaries of marine biotechnology but also provide a replicable model for circular, nature-based economies in saline and arid environments. Strategic interdisciplinary collaboration and supportive policy frameworks are essential to realize the full potential of this approach, enabling a sustainable transition in regions facing acute climate and resource constraints.

Author Contributions

Synan F. AbuQamar: conceptualization, funding acquisition, project administration, supervision, writing – original draft, writing – review and editing. **Khaled A. El-Tarabily:** conceptualization, funding acquisition, writing – review and editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The authors have nothing to report.

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